CS24307 — THEORY (3.0 CR) | MODULE IV NOTES | BIT MESRA | NEP SCHEME

Module 4: Planning & Bayesian Networks

STRIPS, Goal Stack, Probability Axioms, Bayesian Belief Networks

Module IV: Planning & Probabilistic Reasoning — Topics Covered

1. STRIPS Planning Representation & PDDL
2. Goal Stack Planning & Sussman Anomaly
3. Axioms of Probability & Bayes' Rule
4. Bayesian Belief Networks & Joint Factorization
5. D-Separation & Exact Probabilistic Inference

1. Bayesian Network Joint Probability Factorization

$$P(X_1, X_2, \ldots, X_n) = \prod_{i=1}^{n} P(X_i \mid \mathrm{Parents}(X_i))$$

A Bayesian Network is a Directed Acyclic Graph (DAG) whose nodes represent random variables and directed edges represent conditional dependencies.